

軟體測試

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授課時間：每週五 9：10—12：00

博理館 103

課程代號：EE 5170

識別碼：921 U9550

民國 105 年秋季班

期末考

考試中，請勿參閱資料。

1. Below is a graph, defined by the sets of nodes, initial nodes, final nodes, edges, and defs and uses.

Answer the following questions about the graph.

Graph: $N=\{0,1,2,3,4,5,6,7\}$

$N_0=\{0\}$

$N_f=\{6\}$

$E=\{(0,1),(1,2),(2,7),(7,1),(2,3),(2,4),(3,4),(4,5),(5,6),(4,6),(6,2)\}$

$\text{def}(0)=\text{def}(5)=\text{use}(4)=\text{use}(2)=\{x,y\}$

$\text{def}(3)=\text{def}(6)=\text{use}(0)=\text{use}(3)=\{x\}$

$\text{def}(1)=\text{def}(2,4)=\text{def}(2,7)=\text{use}(5)=\text{use}(6)=\{z\}$

All the other define and use sets are empty.

- (a) Draw the graph. (5pts/5)

(b) List the test requirements for edge-pair coverage. (5pts/10)

(c) Please list a set of test paths for full edge-pair coverage. (5pts/15)

(d) List the test requirements for prime-path coverage (PPC). (5pts/20)

(e) Write down a test set for the full PPC. (5pts/25)

(f) List all of the du-paths with respect to x and y . (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/30)

(g) Write down a minimal test set that satisfies all-defs coverage with respect to x and y . (Direct tours only.) (5pts/35)

- (h) Write down a minimal test set that satisfies all-uses coverage with respect to x and y .
(Direct tours only.) (5pts/40)

2. Use the following predicate to answer the following questions.

$$p = (a \vee \neg d) \wedge (\neg a \vee b) \wedge (b \vee c) \wedge (\neg b \vee \neg c \vee d)$$

(a) Identify the clauses (*atoms* in standard logic terms) that determine predicate p .
(5pts/45)

(b) Compute (and simplify) the conditions under which each of the clauses determines predicate p . (5pts/50)

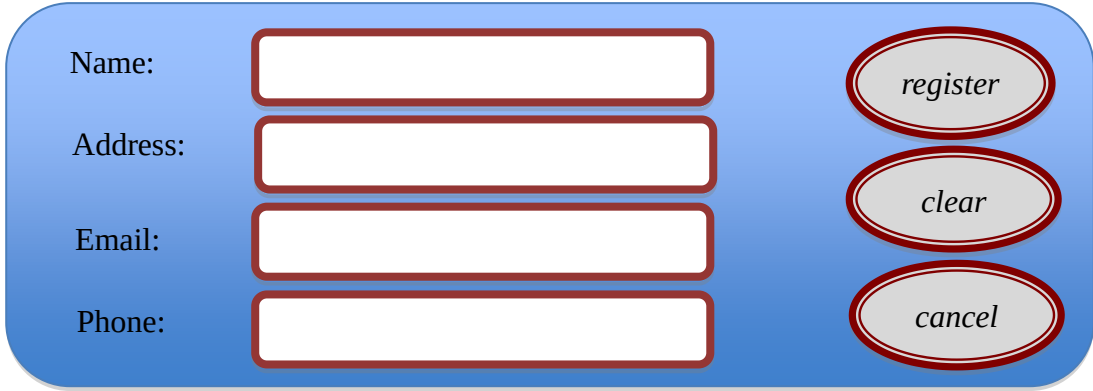
(c) Write the complete truth table for all clauses. Label your rows starting from 0 at the top. Use the format that row 0 should be all clauses true. You should include columns for the conditions under which each clause determines the predicate, and also a column for the predicate itself. (5pts/55)

(d) Identify all pairs of rows from your table that satisfy *general active clause coverage* (GACC) with respect to each clause. (5pts/60)

(e) Identify all pairs of rows from your table that satisfy *correlated active clause coverage (CACC)* with respect to each clause. (5pts/65)

(f) Identify all rows from your table that satisfy *implicant coverage (IC)* with respect to each implicant. (5pts/70)

3. Assume that we have a web program that collect registration data with the following screen layout.



The image shows a registration form with a light blue background and rounded corners. On the left, there are four labels: "Name:", "Address:", "Email:", and "Phone:". To the right of each label is a white rectangular input field with a red border. On the right side of the form, there are three vertically stacked oval buttons with a red border and a light gray background. The buttons are labeled "register", "clear", and "cancel" in a cursive font.

There is no other document. Please answer the following questions to properly test the page.

- (a) Define characteristics of inputs to the web interface and list them. (5pts/75)

(b) Partition the characteristics into blocks and write them down. (5pts/80)

(c) Define and list values for the blocks. (5pts/85)

4. Assume that we have the following program with two mutants.

Original program:

```
public static int search(int[] x, int y) {  
    c = -1;  
    for (int j = 0; j < x.length; j++)  
        if (x[j] == y) c = j;  
    if (c >= 0)  
        return c;  
}
```

- (a) Please write down a test input that kills mutant 1 in the following. Explain why it kills the mutant? (5pts/90)

Mutant 1:

```
public static int count(int[] x, int y) {  
    c = -1;  
    for (int j = 0; j < x.length; j++)  
        if (x[j] = y) c = j;  
    if (c >= 0)  
        return c;  
}
```

(b) Please write down a test input that kills mutant 2 in the following! Please explain why it kills the mutant. (5pts/95)

Mutant 2:

```
public static int search(int[] x, int y) {  
    c = -1;  
    for (int j == 0; j < x.length; j++)  
        if (x[j] == y) c = j;  
    if (c >= 0)  
        return c;  
}
```

5. Please explain how to use language syntax to test compilers. (5pts/100)